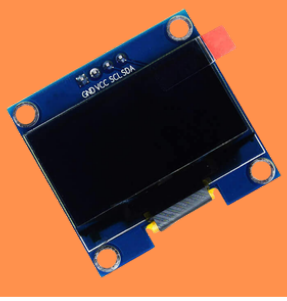
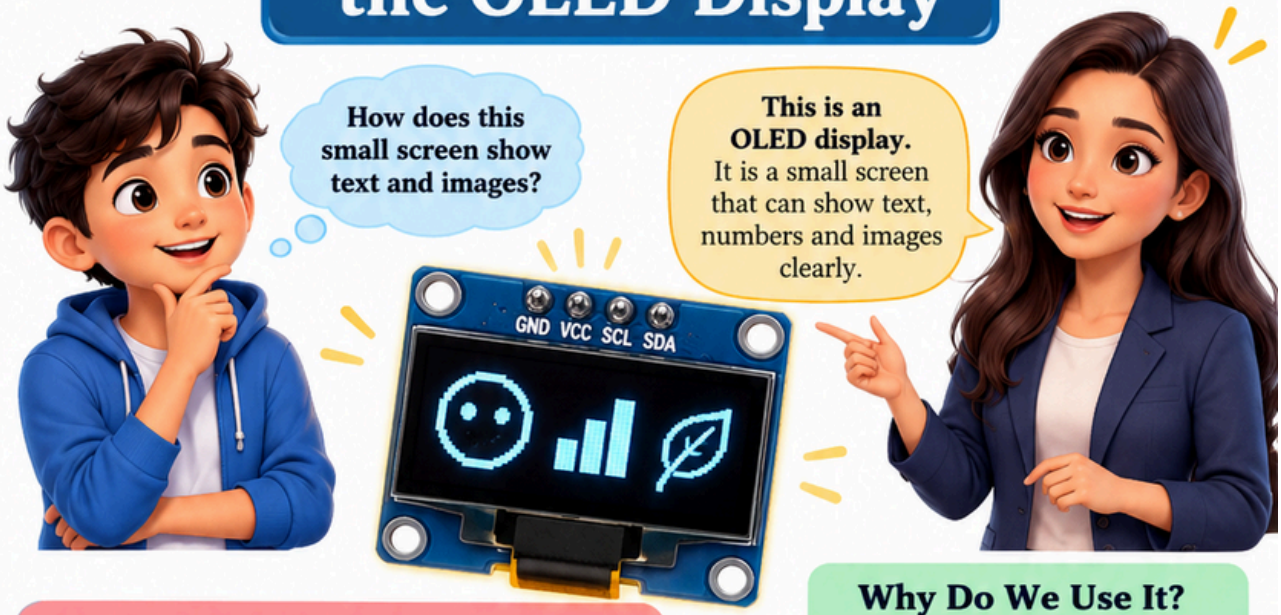
Topic 3

Topic Name: Introduction to OLED Display



Introduction to the OLED Display



What is an OLED Display?

An OLED display is a small and bright screen. It can show text, numbers and simple images. It is used with Arduino to display information from our projects.



Why Do We Use It?

- **Clear Display** — Shows text and images sharply.
- **Small Size** — It is compact and easy to use.
- **Low Power** — It uses less power.
- **Versatile** — It can show text, numbers and simple graphics.

What Can It Show?

Text



Numbers



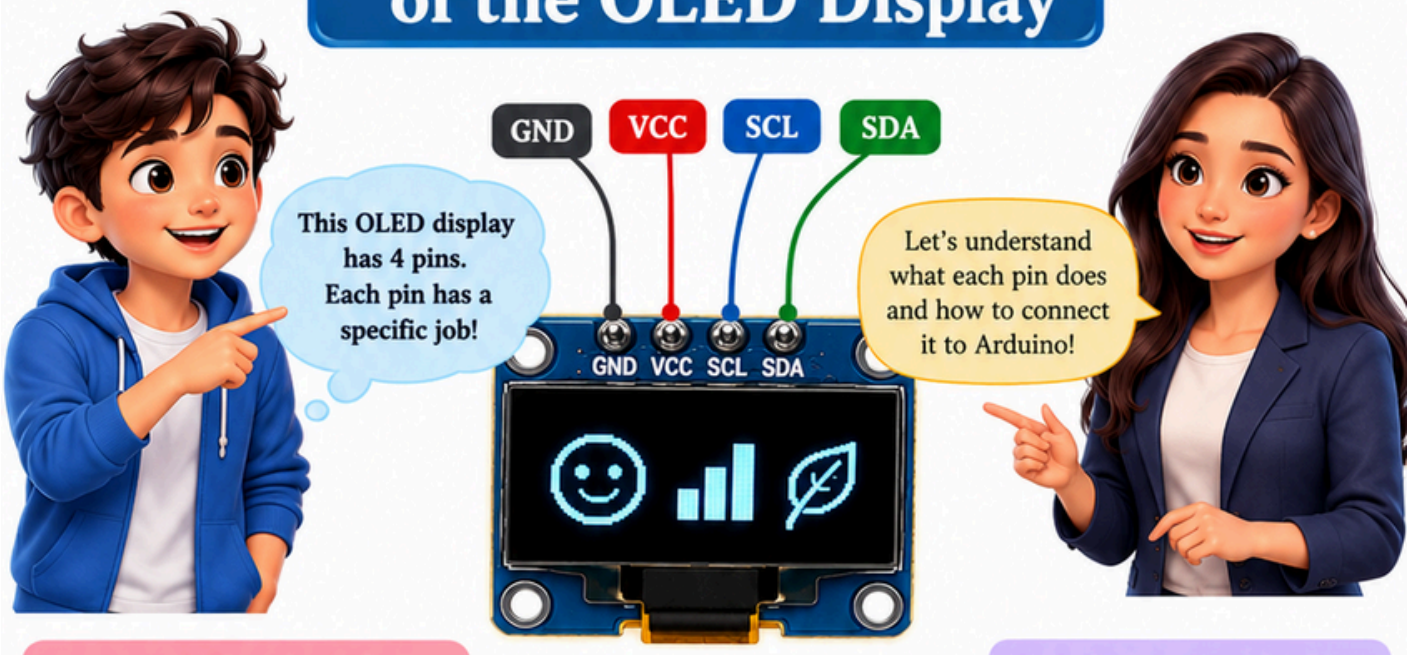
Simple Images



Icons



Pin Configuration of the OLED Display



About Each Pin

GND

Ground pin. It completes the circuit.

VCC

Power pin. It provides 5V or 3.3V power to the display.

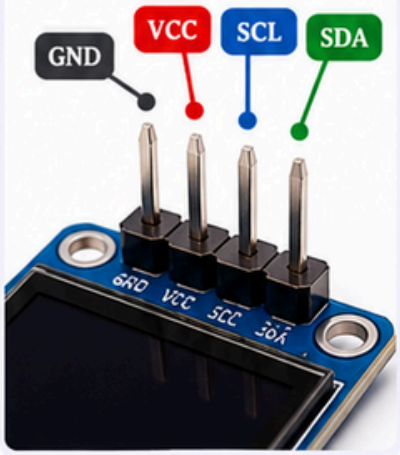
SCL

Serial Clock pin. It helps to send clock signals for communication.

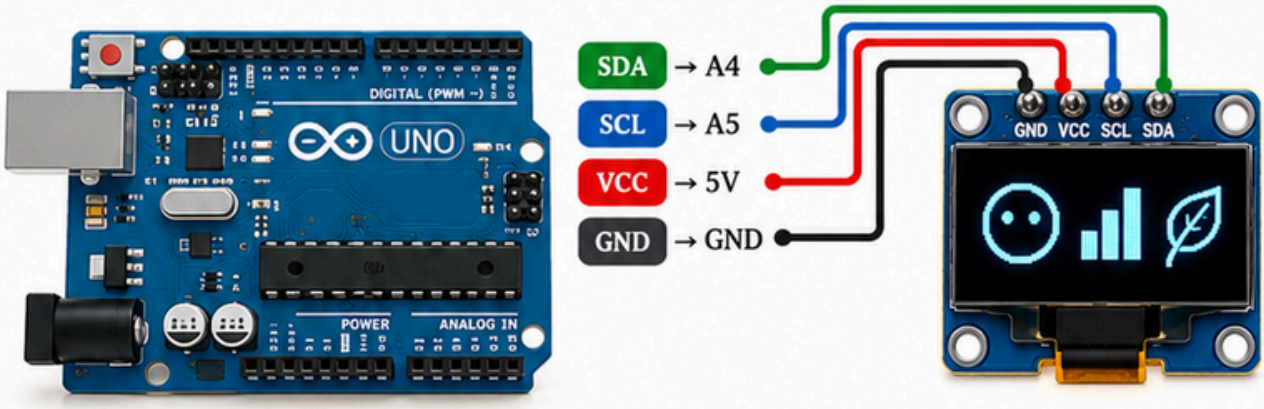
SDA

Serial Data pin. It helps to send and receive data between Arduino and the display.

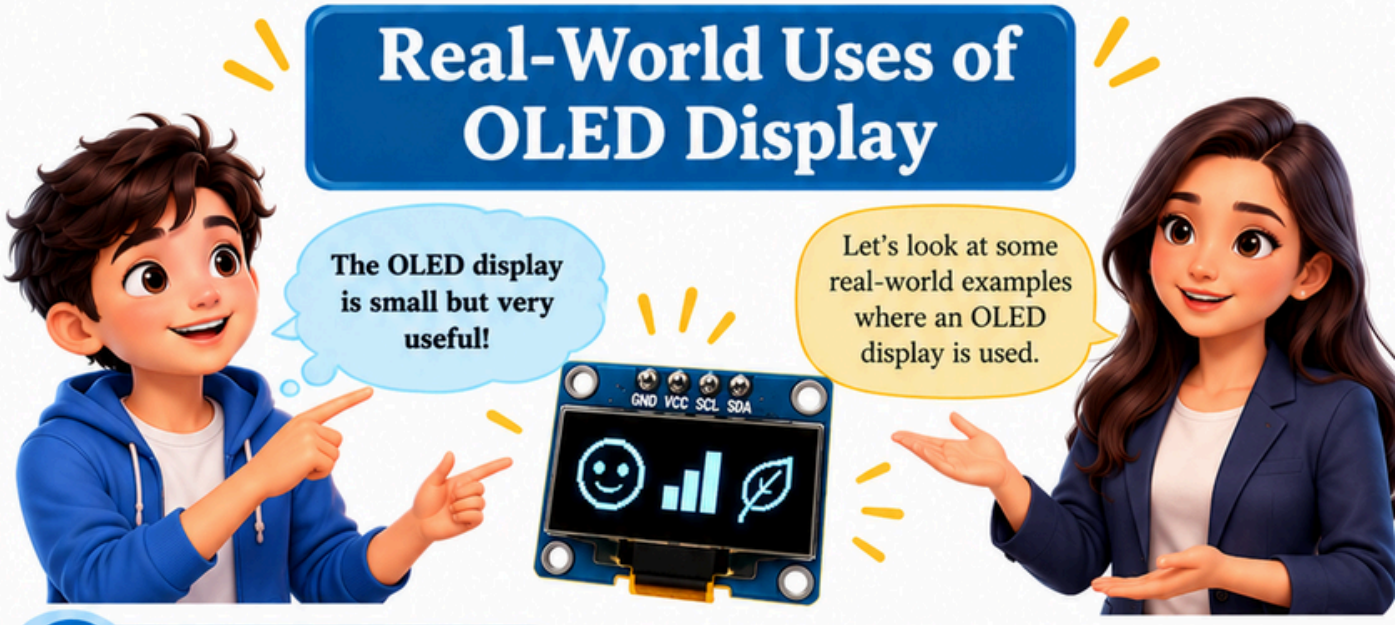
Side View of Pins



Connecting OLED Display to Arduino



Real-World Uses of OLED Display



1 Smart Watch

OLED displays are widely used in smart watches. They can show the time, steps, heart rate and notifications clearly, even in bright sunlight.



What it Shows

- Time
- Steps
- Heart rate
- Notifications



2 Digital Thermometer

OLED displays are used in digital thermometers to show temperature readings clearly. The display is bright and easy to read, even from different angles.



What it Shows

- Temperature value
- Units (°C or °F)
- Battery level
- Some icons



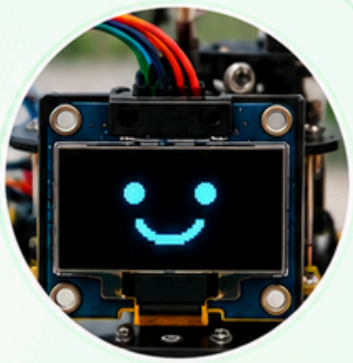
3 Robotics and DIY Projects

OLED displays are commonly used in robotics and DIY projects. They can show sensor data, messages and simple graphics in a small space.



What it Shows

- Sensor readings
- Messages
- Icons and shapes
- Project status



Learning Outcomes

- The student will be able to identify an OLED display and explain what it is used for.
- The student will be able to identify the four OLED pins: GND, VCC, SCL, and SDA.
- The student will be able to explain how an OLED displays information such as text, numbers, and simple images.
- The student will be able to identify real-world uses of OLED displays in devices like smartwatches, thermometers, and robots.